

KALAIYARASAN M

Coimbatore, Tamil Nadu, India | +91 9940982901 | kalaiyarsan.m.dev@gmail.com | github.com/kalaiyarsan-sys

PROFESSIONAL SUMMARY

Python software engineer with 4+ years of backend development experience across ERP/CRM systems, REST APIs, and relational databases, including CI/CD pipeline design and automated testing that measurably improved deployment speed and release quality. Open-source contributor with a merged upstream fix to Frappe/ERPNext, including root-cause debugging, environment setup, and the full fork-to-merge pull request workflow. Currently building hands-on experience with LLM applications, Retrieval-Augmented Generation, and AI evaluation — including direct use of the Ragas framework for evaluating RAG pipelines — and studying agentic AI and Model Context Protocol (MCP) concepts. Looking to apply a Python engineering and open-source contribution background to developer tools and evaluation infrastructure for LLM applications.

PROFESSIONAL EXPERIENCE

Software Engineer, Python Web Developer — Digixr Technologies

Jan 2022 – Present

- Developed and maintained backend applications on the Frappe Framework and ERPNext ecosystem, including ERP/CRM modules.
 - Built and integrated REST APIs; designed and managed relational database schemas (MySQL/MariaDB).
 - Debugged and resolved production software issues, working from bug reports through root-cause analysis to a tested fix.
 - Designed and built an automated test suite that reduced manual QA effort by 30%, improving release confidence.
 - Redesigning the deployment pipeline using Bitbucket Pipelines to automate pushes to production and pilot servers, cutting deployment time by 20% and improving system stability.
 - Tested applications, optimized performance, and supported deployments while working from business requirements.
 - Wrote maintainable, reviewable code as part of a small engineering team.
-

OPEN SOURCE CONTRIBUTION

Frappe / ERPNext — Open-Source Contributor

Issue #54022 | github.com/frappe/erpnext/issues/54022

- Diagnosed a Sales Order status bug in ERPNext where fully invoiced and paid non-stock orders were incorrectly marked “To Deliver” or “Overdue.”
 - Set up a local Frappe/ERPNext development environment using Docker and the Bench CLI, and reproduced the issue end-to-end.
 - Forked the upstream repository, traced the issue through the codebase, and implemented and tested a fix.
 - Opened a pull request to the upstream frappe/erpnext repository; the fix was reviewed and merged upstream.
-

CURRENT TECHNICAL FOCUS — LLM & AI EVALUATION

- Actively learning LLM application development, Retrieval-Augmented Generation (RAG), and AI agent / multi-agent system design.
 - Hands-on with the Ragas evaluation framework: ran a local TF-IDF + Ragas evaluation pipeline and worked directly with non-LLM metrics (BleuScore, RougeScore, NonLLMStringSimilarity, ExactMatch).
 - Exploring LangChain and LangGraph for building LLM applications, and studying LLM evaluation, observability, and reliability concepts.
 - Studying the Model Context Protocol (MCP) and how it standardizes tool/context access for LLM applications, as part of broader agentic-AI research.
 - Transitioning a Python engineering and open-source contribution background toward LLM developer tools and evaluation infrastructure.
-

PROJECTS

Farmer’s Market Web Application — Personal Project

- Built a Flask backend with authentication, product management, order processing, and database operations.

ML-Based Lifestyle Prediction, Post-COVID — Personal Project

- Applied regression, decision trees, and neural networks to analyze post-COVID lifestyle data.
-

TECHNICAL SKILLS

Programming: Python, JavaScript, SQL

Backend & Frameworks: Frappe Framework, ERPNext, Django, Flask, REST APIs

Databases: MySQL, MariaDB

CI/CD & DevOps: Bitbucket Pipelines, Docker, Git, GitHub, Bitbucket

Developer Tools & Open Source: Git, GitHub, Docker, Bench CLI, Pull Requests, Code Review

Engineering: Debugging, Automated Testing, Performance Optimization, Deployment, Problem Solving

Current AI Focus: LLMs, RAG, AI Agents, Ragas, LangChain, LangGraph, MCP (studying), AI Evaluation Concepts

EDUCATION

B.Tech in Information Technology — SNS College of Engineering | CGPA: 7.79